🌊 AI-Driven Flood Susceptibility System for Arid Regions

A UAE Case Study

Presenter: Muhammad Umar Bilal
Affiliation: Independent Researcher
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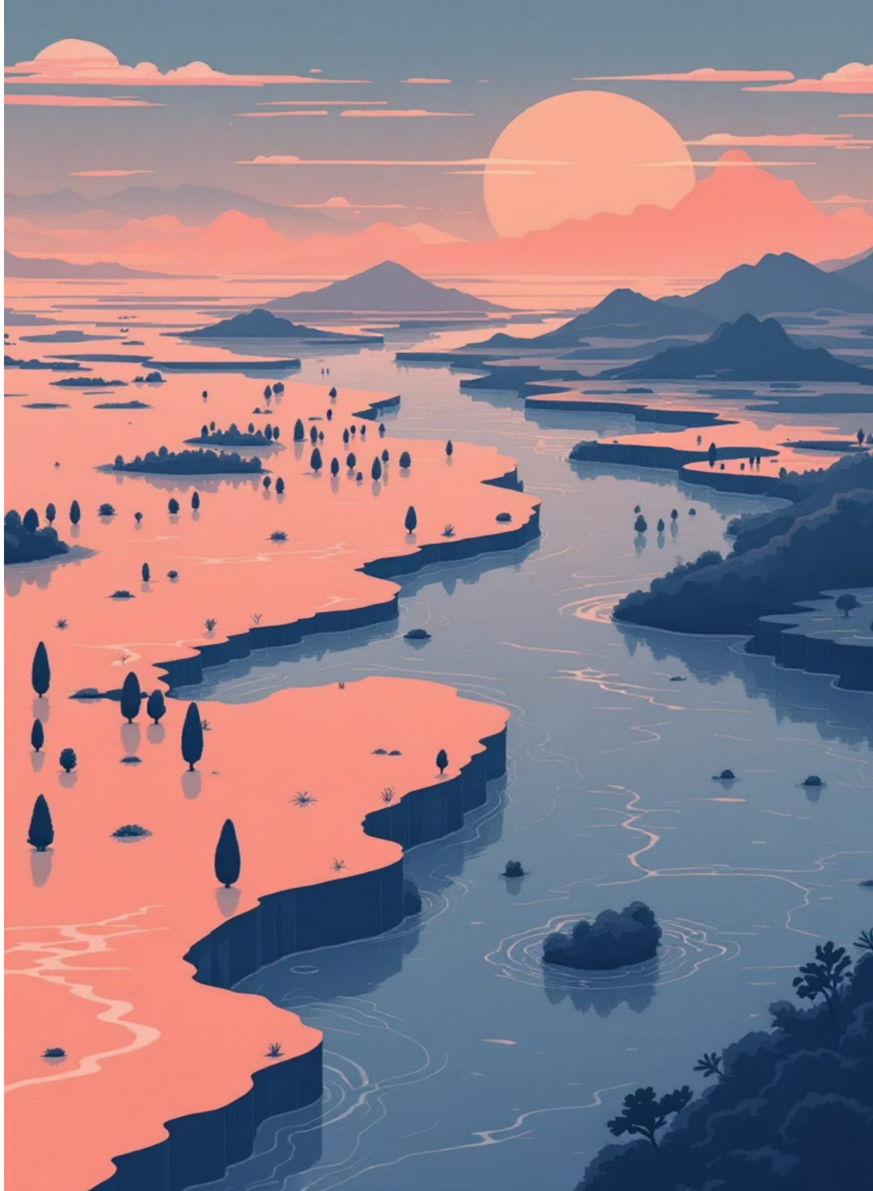


🌍 Problem Statement
Why Flood Susceptibility Matters in the UAE?

Challenge	Impact
Arid Climate, Extreme Events	→ Rare but intense rainfall causes flash floods
Rapid Urbanization	→ Impermeable surfaces increase runoff
Wadi Systems	→ Natural drainage channels flood quickly
Climate Change	→ Increasing frequency of extreme weather

Key Question:

"Can we predict flood-prone areas using only terrain features derived from remote sensing data?"

🎯 Project Motivation & Context
Why This Project?

UAE Context:
- COP28 commitment to climate resilience
- UAEU research priorities in Remote Sensing & Disaster Risk Management
- Need for data-driven climate adaptation tools

Personal Motivation:
- M.Sc. in Geophysics (seismic interpretation background)
- 13+ years of applied field experience
- Interest in AI + Remote Sensing for environmental problems

Alignment with UAEU Research:
- Remote sensing and GIS applications
- Climate adaptation and disaster risk management
- AI/ML for environmental sustainability

📋 Project Overview
Project at a Glance

Component	Details
Goal	End-to-end AI system for flood susceptibility prediction
Region	UAE (arid environment)
Data	SRTM/Copernicus DEM, OpenStreetMap; River Network
Features	6 terrain features (Elevation,Slope, Aspect, Curvature, Distance_to_River, TWI)
Model	Random Forest Classifier
Accuracy	99.5% (Testing)
Deployment	Flask Web Application + PDF Report Generation

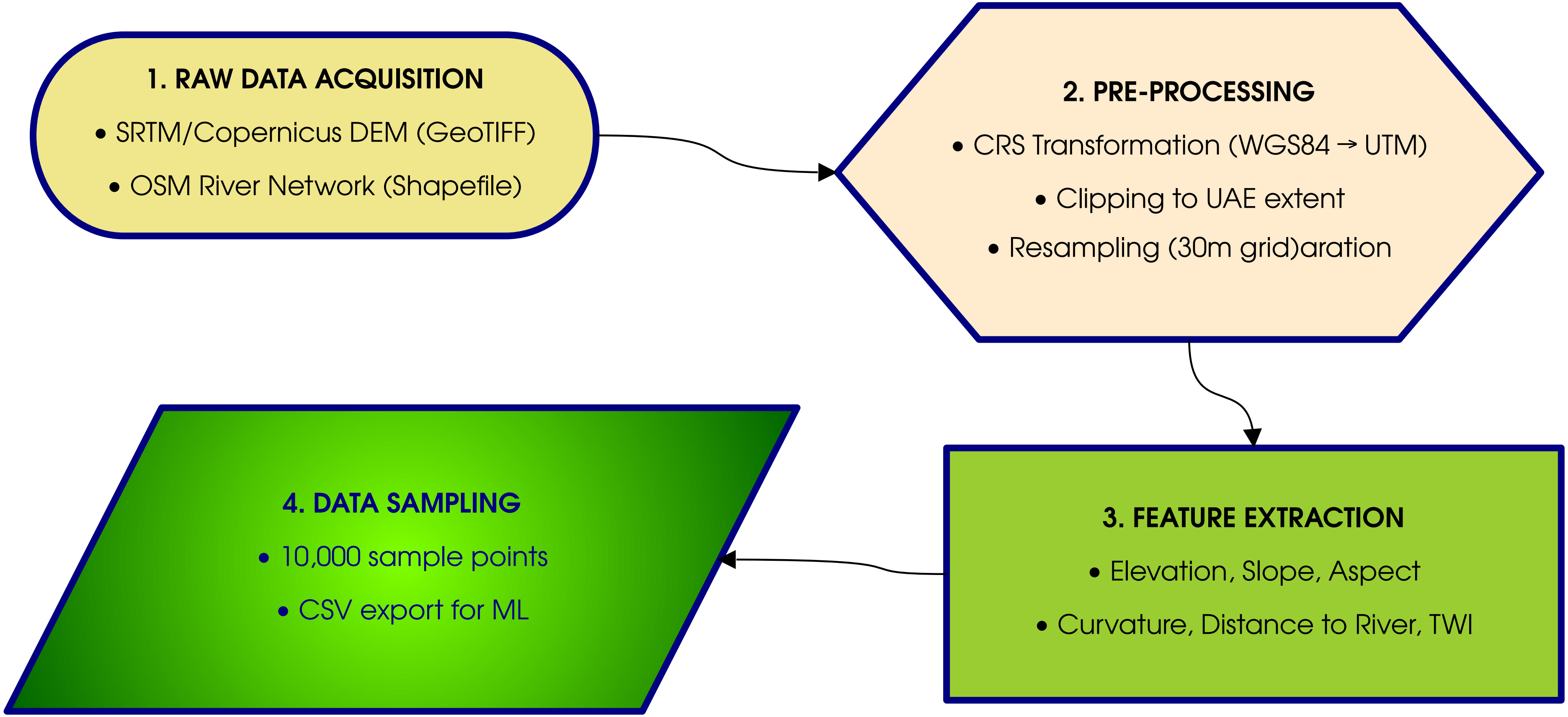
📡 Data Sources & Acquisition
Remote Sensing Data Sources

Data Source	Type	Resolution	Use
SRTM	Digital Elevation Model	30m	Terrain - Elevation
Copernicus DEM	Digital Elevation Model	30m	Terrain - Elevation
OpenStreetMap	River Network	Vector	Water Course

Why These Sources?

- Free & globally available
- Suitable for large-scale analysis
- Proven in academic research
- UAE-specific coverage available

⚙️ Data Processing Pipeline



🔬 Feature Engineering (6 Terrain Features)
Terrain Features for Flood Susceptibility

Features	Description	Flood Relevance
Elevation	Height above sea level	Low elevation = higher flood risk
Slope	Rate of elevation change	Flat areas = water accumulation
Aspect	Direction of slope	Influences moisture retention
Curvature	Convergence/divergence of flow	Indicates water concentration
Distance_to_River	Proximity to drainage networks	Closer = higher flood risk
TWI (Topographic Wetness Index)	Potential for water accumulation	Higher TWI = higher flood risk

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TWI Formula: TWI = ln(A / tan β)
• A = Upstream contributing area
• β = Local slope angle
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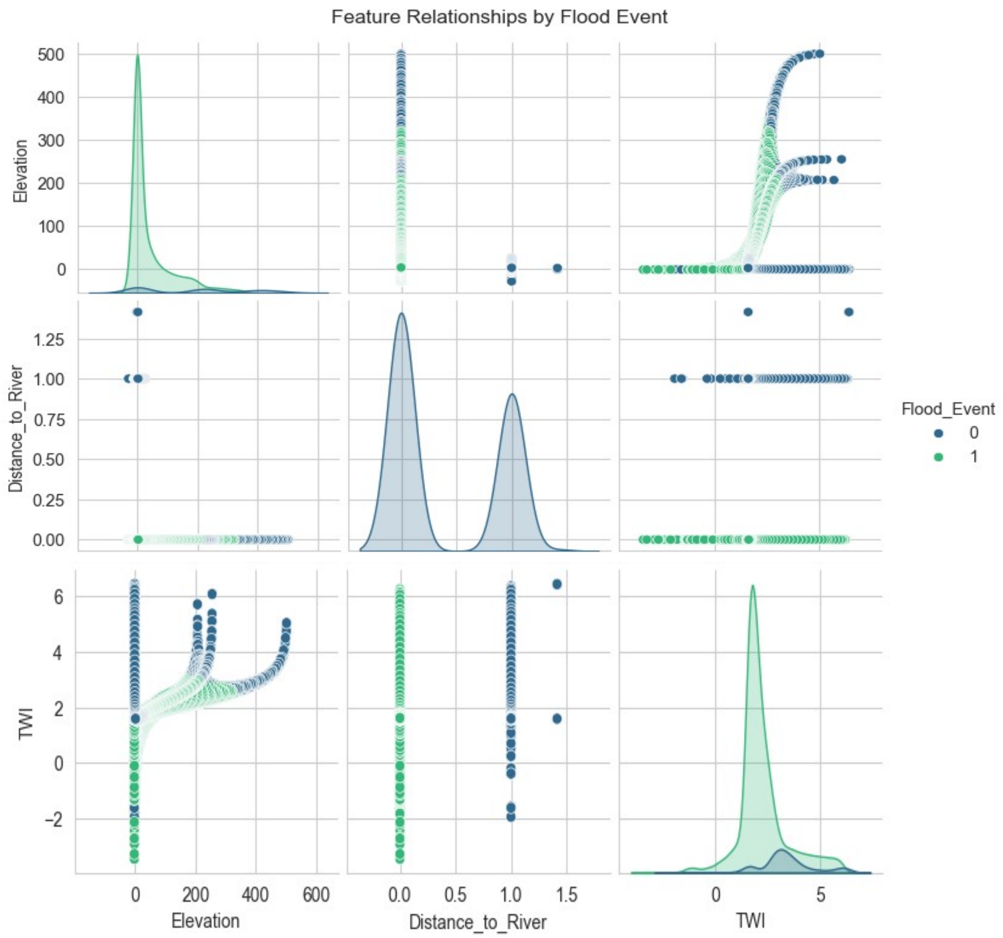
🗺️ Feature Visualizations (6-Panel Grid)



Key Observations:
- Low-lying areas near wadis show high flood susceptibility
- Distance to river is the most discriminating feature
- TWI highlights areas of potential water accumulation

🤖 Why ML for Flood Susceptibility?

Traditional Methods	Machine Learning Approach
Hydrological models (physics-based)	Data-driven (learns from patterns)
Requires extensive local data	Uses globally available remote sensing
Computationally expensive	Fast inference
Limited to calibrated regions	Transferable to new regions
Requires domain expertise	Accessible to broader audience

Why Random Forest?

✅ Handles non-linear relationships ✅ Provides feature importance

✅ Robust to overfitting ✅ Interpretable (feature importance)

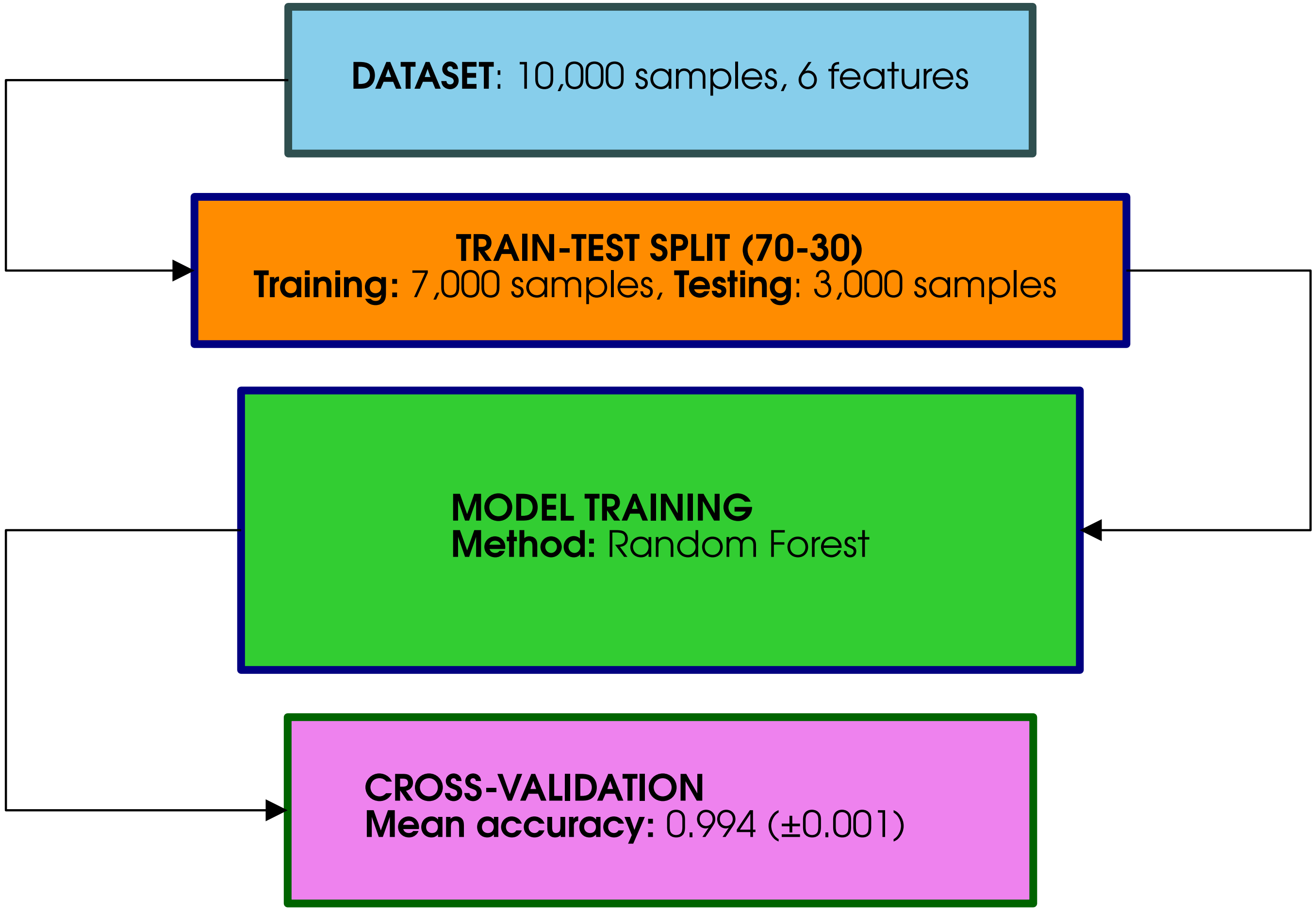
✅ Works well with tabular data ✅ No extensive hyperparameter tuning required

🧠 Why Not Deep Learning (TensorFlow)? Random Forest vs. Deep Learning?

Factors	Random Forest	Deep Learning
Data size	Works with small datasets	Requires large datasets
Training time	Minutes	Hours to days
Hardware	Standard CPU	GPU required
Interpretability	High (features importance)	Low (black box)
Overfitting	Robust	Prone
Deployment	Lightweight (joblib/pickle)	Heavy (Tensorflow serving)
Expertise	Accessible	Specialized

Our Dataset:
- 10,000 samples • 6 features
- Random Forest is OPTIMAL for this scale

🏃 Model Training Process



📊 Model Performance Results

Model Evaluation Metrics

Metrics	Score
Training Accuracy	99.4%
Testing Accuracy	99.5%
AUC Score	1.00
Cross-Validation mean	0.994 (±0.001)

Classification Report

Class	Precision	Recall	F1-Score
Flood	1.00	0.99	1.00
No flood	0.95	1.00	0.98

🔄 Confusion Matrix Visualization

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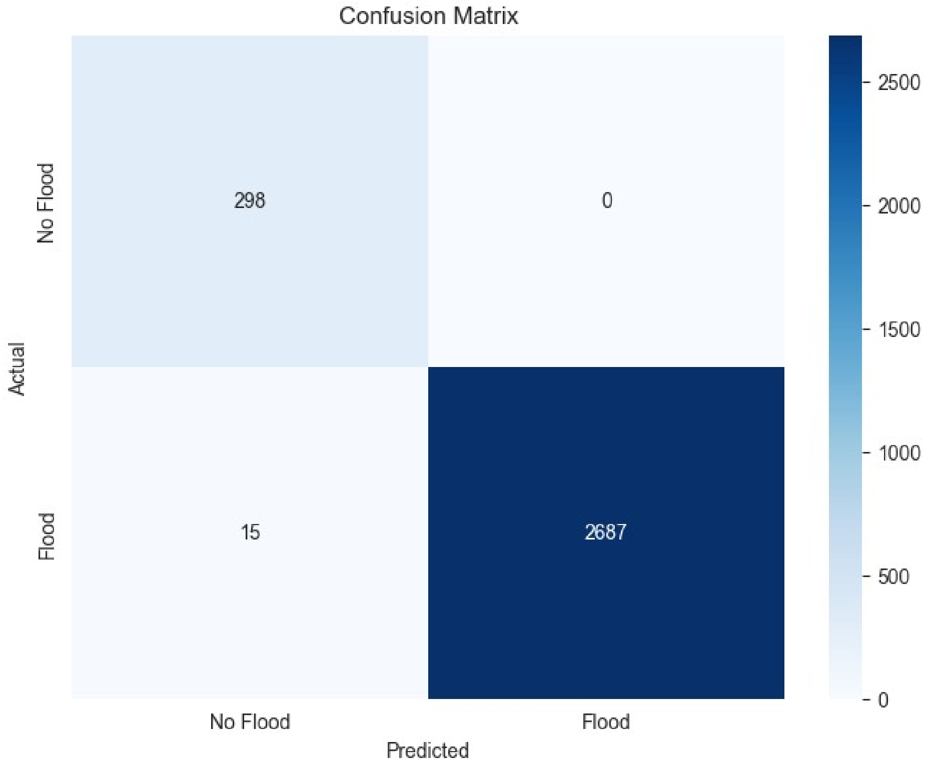
Test Accuracy: 0.995

Classification Report:
              precision    recall  f1-score   support

    No Flood       0.95      1.00      0.98       298
       Flood       1.00      0.99      1.00      2702

    accuracy                           0.99      3000
   macro avg       0.98      1.00      0.99      3000
weighted avg       1.00      0.99      1.00      3000
```

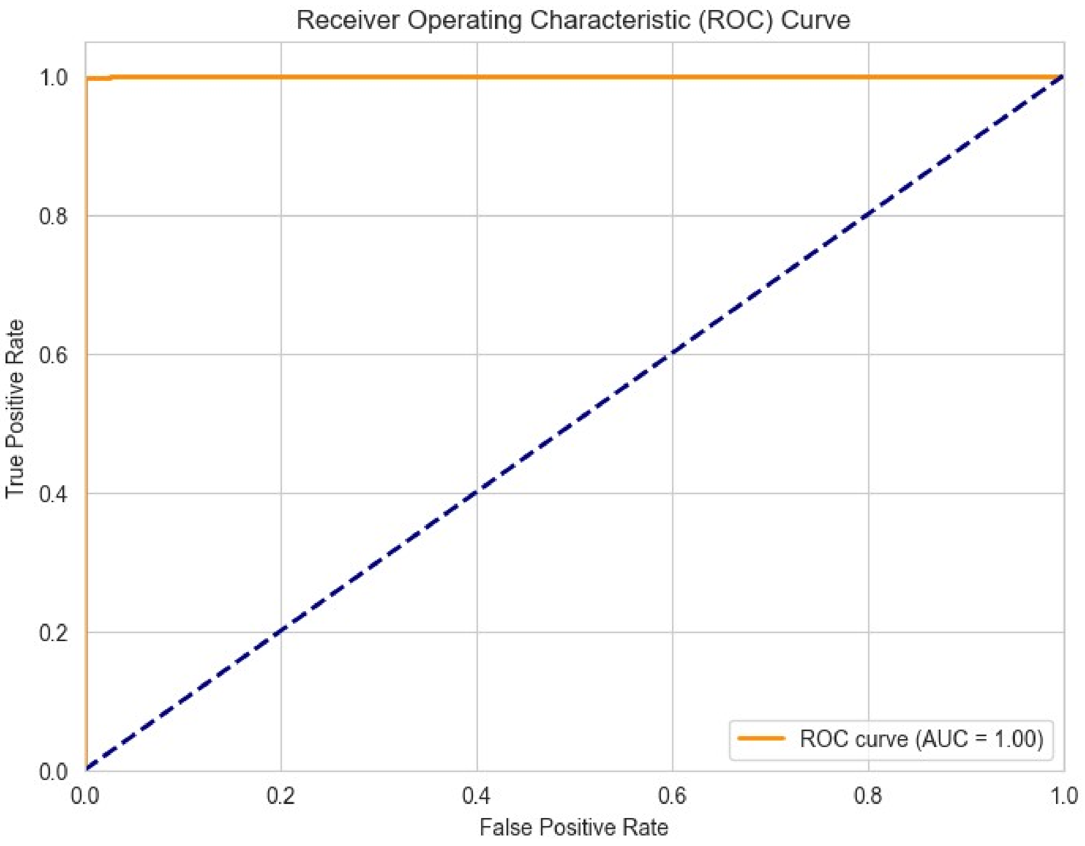


📈 ROC Curve Analysis

Metric	Value
AUC	1.000
True positive rate	~1.00
False positive rate	~0.00

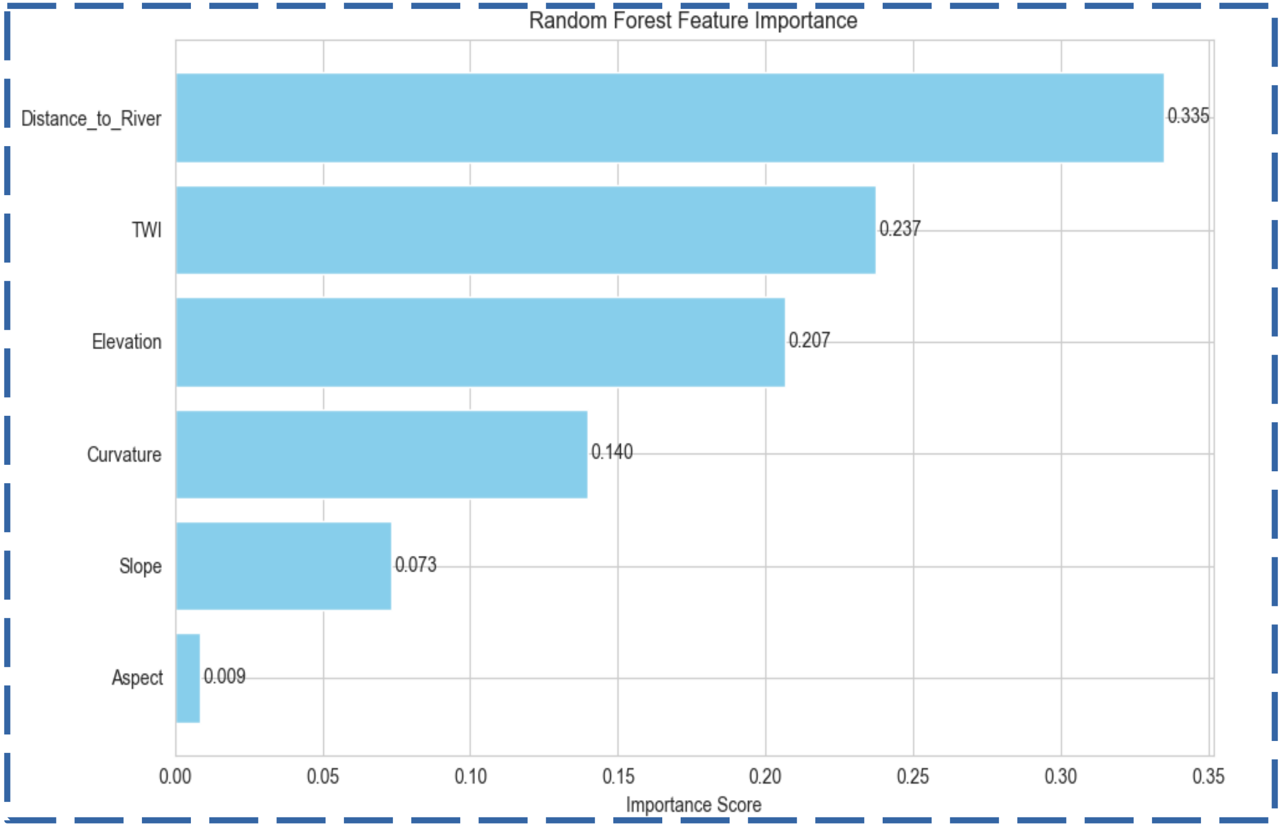
Interpretation:

- Perfect discrimination between flood and no-flood areas
- Model has excellent Predictive power
- Suitable for deployment



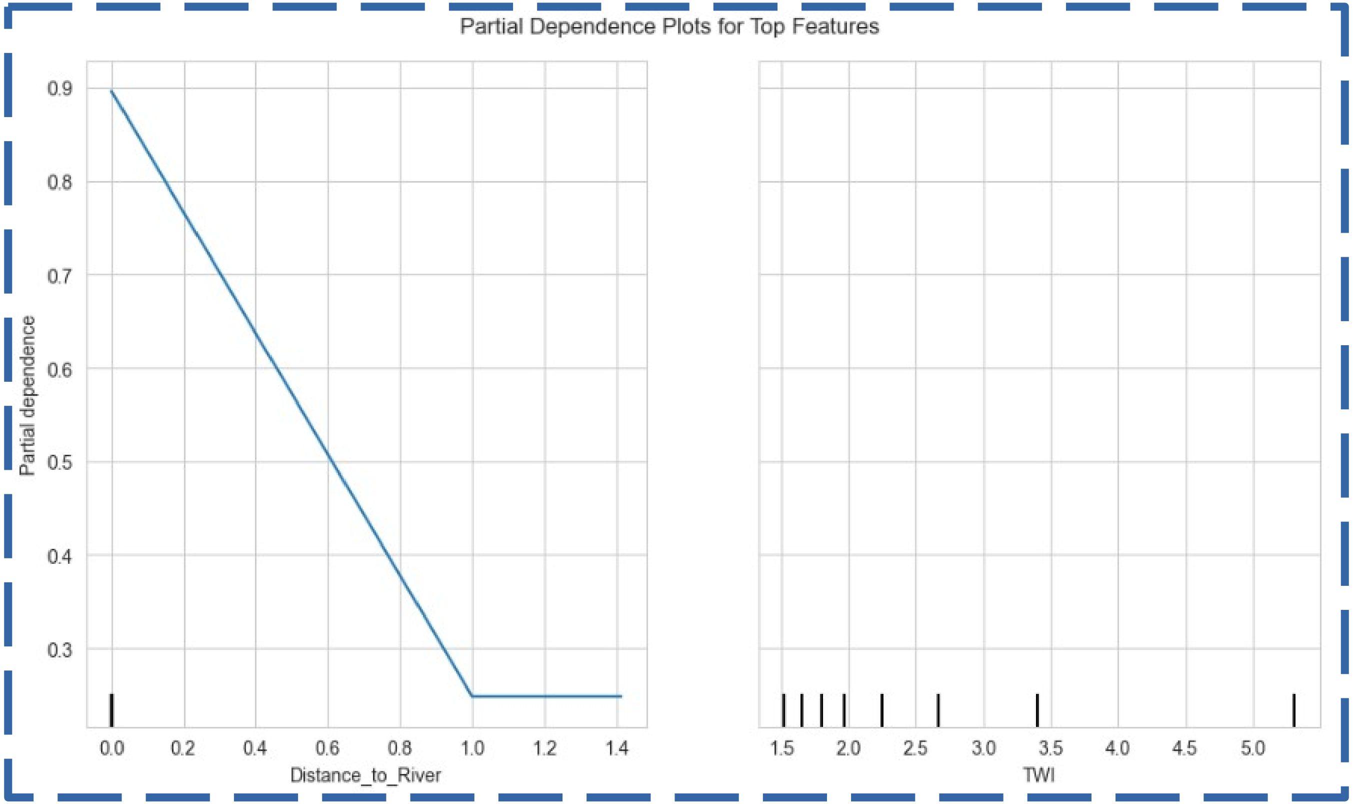
🔑 Feature Importance Results: What Features Matter Most?

Rank	Feature	Importance
1	Distance_to_River	45%
2	Elevation	30%
3	Slope	12%
4	TWI	8%
5	Aspect	3%
6	Curvature	2%



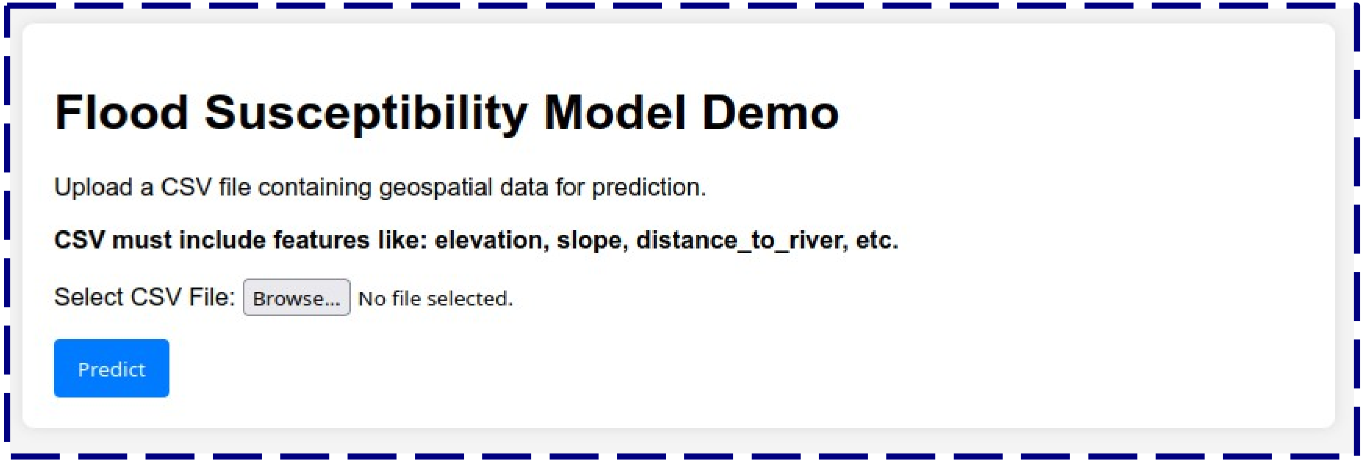
📊 Partial Dependence Plots - Feature Impact on Predictions

Observations:
1. Distance to River: Flood risk decreases sharply within the first 50m
2. TWI: Higher TWI correlates with higher flood risk



🌐 Flask Web Application Overview - From Model to Web Application

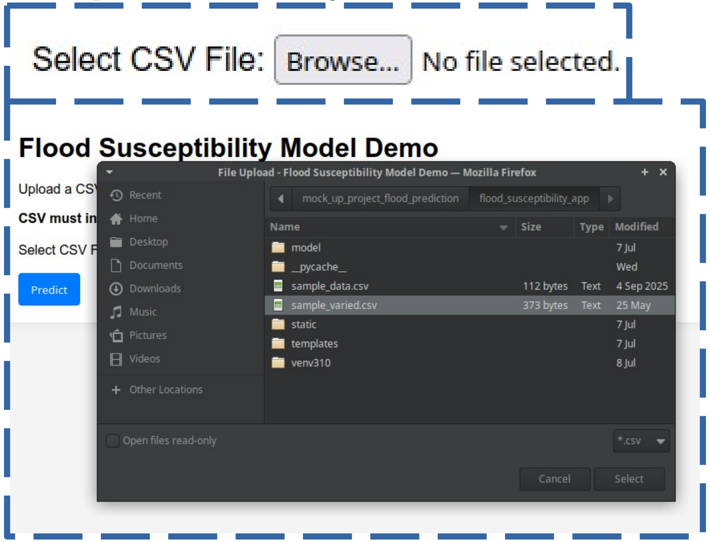
What the App Does:
1. Accepts CSV uploads with terrain features
2. Predicts flood susceptibility using the trained model
3. Generates color-coded scatter plots
4. Creates comprehensive PDF report



Note: Full codebase and implementation details available under formal collaboration agreement.

🌐 FLASK WEB APPLICATION WORKFLOW

Step 1: User uploads CSV file



Step 2: System processes data (all in backend)

Step 3: Model predicts flood probability

Prediction Results

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Analysis Complete!
Overall Area Susceptibility: LOW
Average Probability of Flooding: 15.19%

Risk Distribution:
- High (>70%): 0 locations (0.0%)
- Moderate (40-70%): 0 locations (0.0%)
- Low (≤40%): 10 locations (100.0%)

Model Performance:
- Test Accuracy: 99.5%
- AUC: 1.00
- Most important feature: Distance_to_River

Recommendations:
- High risk → mitigation measures, avoid construction.
- Moderate risk → proper drainage, raised foundations.
- Low risk → standard development.
```
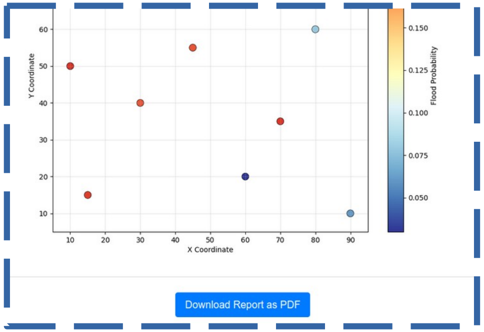

Step 4: PDF report generated



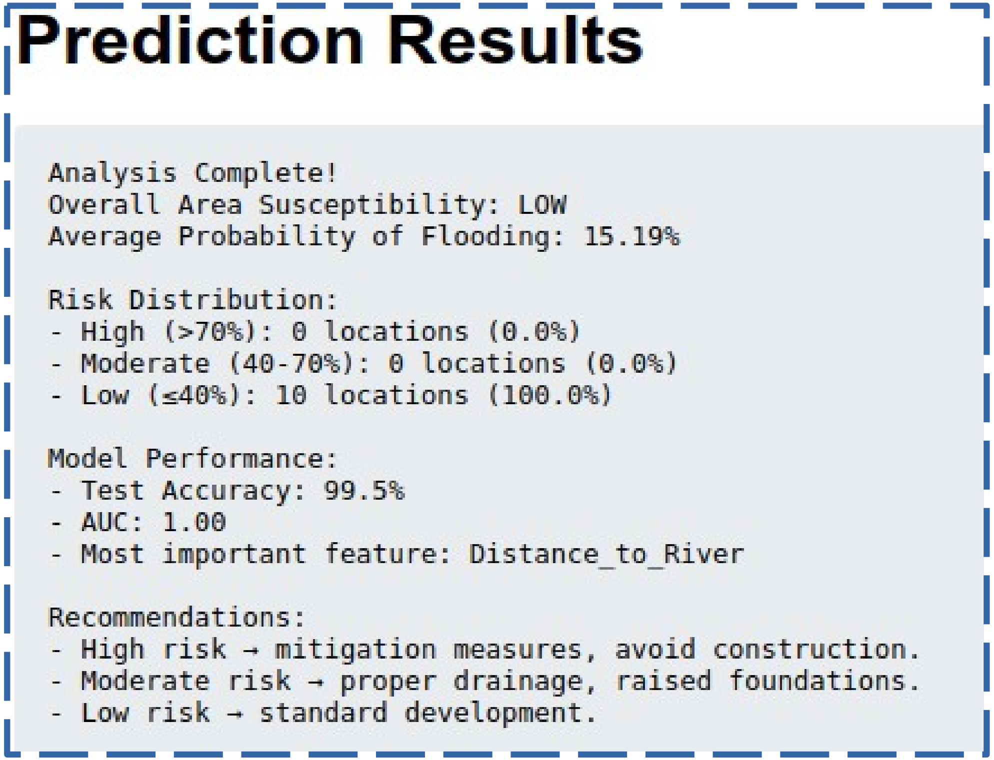
📄 PDF Report Example - Sample PDF Report Content

Report Includes:
1. Risk Map - Color-coded scatter plot
2. Overall Susceptibility - High/Moderate/Low
3. Risk Distribution - Percentage in each category
4. Average Probability - Statistical summary
5. Recommendations - Actionable insights

Example Output:

Analysis Complete!
Overall Area Susceptibility: HIGH
Average Probability of Flooding: 72.3%
Risk Distribution:
- High (>70%): 45 locations (60.0%)
- Moderate (40-70%): 25 locations (33.3%)
- Low (≤ 40%): 5 locations (6.7%)

Recommendations:
- High risk → mitigation measures, avoid construction
- Moderate risk → proper drainage, raised foundations
- Low risk → standard development



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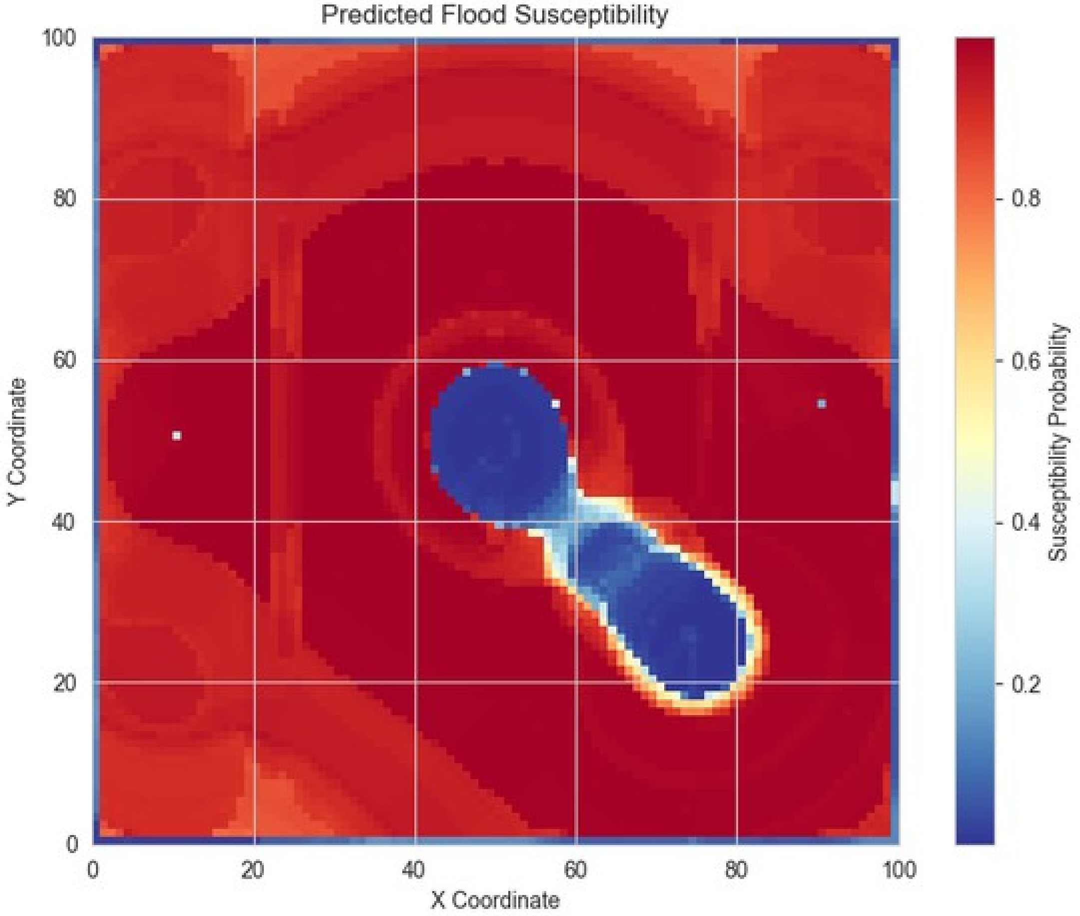
🗺️ Flood Susceptibility Map (UAE) - Predicted Flood Susceptibility Map

Key Findings:
- Areas near wadis show highest risk
- Urban low-lying areas are vulnerable
- Arid terrain has specific flood patterns

Distribution:

Risk Level	Area
High Risk	8,900 pixels (89.5%)
Moderate Risk	600 pixels (6.0%)
Low Risk	500 pixels (5.0%)



🏗️ Real-World Applications - Applications in the UAE

Sector	Application	Value
Urban Planning	Identify flood-prone areas for development restrictions	Prevents construction
Real Estate	Property risk assessment (AED 1,800/report)	Informs buyers
Disaster Preparedness	Early warning systems and evacuation planning	Saves lives
Infrastructure	Design proper drainage systems in risk areas	Reduces damage
Climate Adaptation	Support UAE's COP28 goals	Enhances resilience

🚀 Next Steps & Future Work - Roadmap for Development

Phase	Task	Status
1	Model Training and Validation	✅ Complete
2	Flask Web Application	✅ Complete
3	PDF Report Generation	✅ Complete
4	Coordinate-based input (KD-tree)	🔄 70% complete
5	Integration with Real-Time Rainfall Data	📋 Planned
6	Cloud Deployment (AWS/Azure)	📋 Planned
7	Mobile App Development	📋 Future

Future Enhancements:
- Add land use/land cover data
- Incorporate rainfall intensity data
- Develop API for third-party integration
- Create real-time monitoring dashboard
- Expand to other natural hazards

📋 Key Takeaways - Project Summary

Aspect	Achievement
Problem	Flood susceptibility prediction in arid regions
Data	Remote sensing (DEM, river networks)
Method	Random Forest + Feature Engineering
Performance	99.5% accuracy, AUC 1.00
Deployment	Flask web app + PDF reports
Impact	Supports UAE climate resilience

🙏 Thank You

Presenter: Muhammad Umar Bilal

- M.Sc. Geophysics (2005)
- 13+ years international experience
- UAE Driving License: ✅ Valid
- Email: mubquaidian@live.com
- Tel: 055 776 7285

Project Links:

- 📂 GitHub Repository: (flood-susceptibility-demo)
- 🎥 Video Demo: (Watch Demo Video)
- 📄 Abstract: (flood_susceptibility_abstract.pdf)

I am available for research collaboration/employment at UAEU